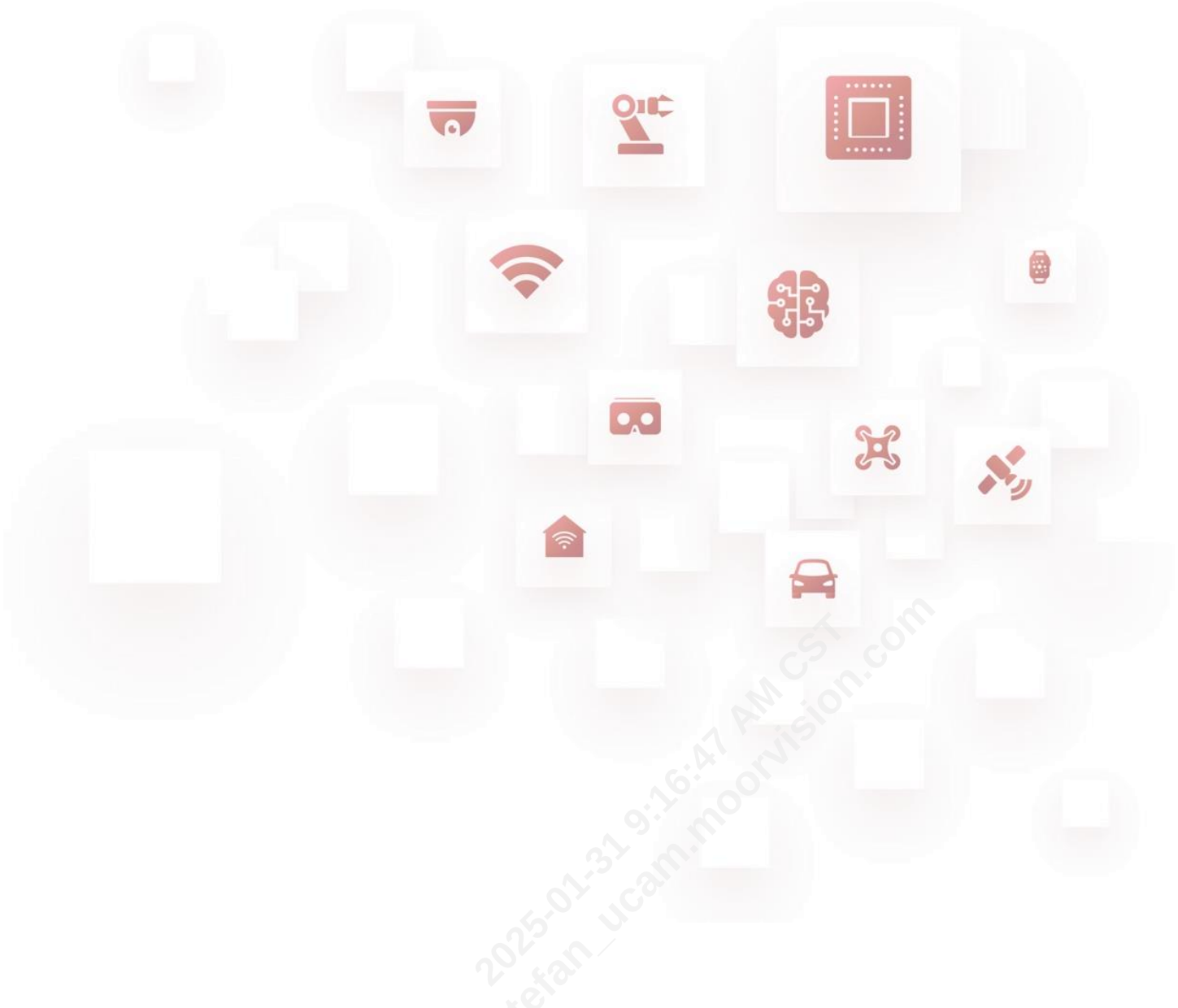
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Thundercomm TurboX C410/C610 Open Kit Linux Android

Software Release Notes

Rev. V1.0
Sep 22, 2023

DN: tc-gla-2410
SDK Version: LA2.0.Post-CS2.r002003

Revision History

Revision	Date	Author	Description
1.0	Sep 22, 2023	Ranran Cong	Initial release.

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About This Document

- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Introduction

This document provides information on how to obtain, build, and program QCS410/610 LA (Linux Android) SDK (Software Development Kit) into the TurboX C410/C610 Open Kit.

➲ **NOTE:** "TurboX" referred to herein is the English text of our registered trademark TURBOX.

The Linux Android Software Development Kit supports the Android operating system on the Thundercomm TurboX. The goal of the LA SDK is to provide a simple and flexible development platform to rapidly build IoT products with the QTI chipsets. The LA SDK provides the build framework, software components, tools, and sample applications to evaluate the platform and build software images for the target product.

➲ **NOTE:** This release only supports the 4GB TurboX C610/C410 Open Kit. Running it on a 2GB version would cause certain functions to be unavailable due to insufficient memory.

Table 1-1. TurboX C410/C610 platform specifications

SOC	SKUx		Standard SKU[1)]	OS	Thundercomm Product No. (TPN)	Model
QCS610	SKU1	2GB + 16GB	Yes	LA[2)]	C610-D21-369PWXX C610-D21-ACXPW00 (LA Specified)	TurboX C610
	SKU2	2GB + 64GB	No	LA[2)]	C610-D26-369PWXX C610-D26-ACXPW00 (LA Specified)	TurboX C610
	SKU3	4GB + 64GB	Yes	LA	C610-D46-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C610
QCS410	SKU4	2GB + 16GB	Yes	LA[2)]	C410-D21-369PWXX C610-D46-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU5	2GB + 64GB	No	LA[2)]	C410-D26-369PWXX C410-D26-ACXPW00 (LA Specified)	TurboX C410
QCS410	SKU7	4GB + 64GB	No	LA	C410-D46-369PWXX C410-D46-ACXPW00(LA Specified)	TurboX C410

➲ **NOTES:**

[1)] indicates standard SKU series. For standard SKU series, retail stock is available; as to non-standard SKU series, in case of any customer requirement, specific orders need to be placed accordingly.

[2)] indicates that when the memory is less than 4GB, Android 12 OS runs out of memory. In this case, it is necessary to tailor the system as required to stabilize the system.

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Chapter 2. Development Environment Setup

Refer to *TurboX SOM SDK Manager User Guide* to set up the development environment.

➲ **NOTE:** Make sure that you use the sdkmanager built based on Ubuntu 18.04 with the "`bash turbox-sdkmanager-setup.sh --os-version 18.04 -u`" command, as shown in Figure 2-1:

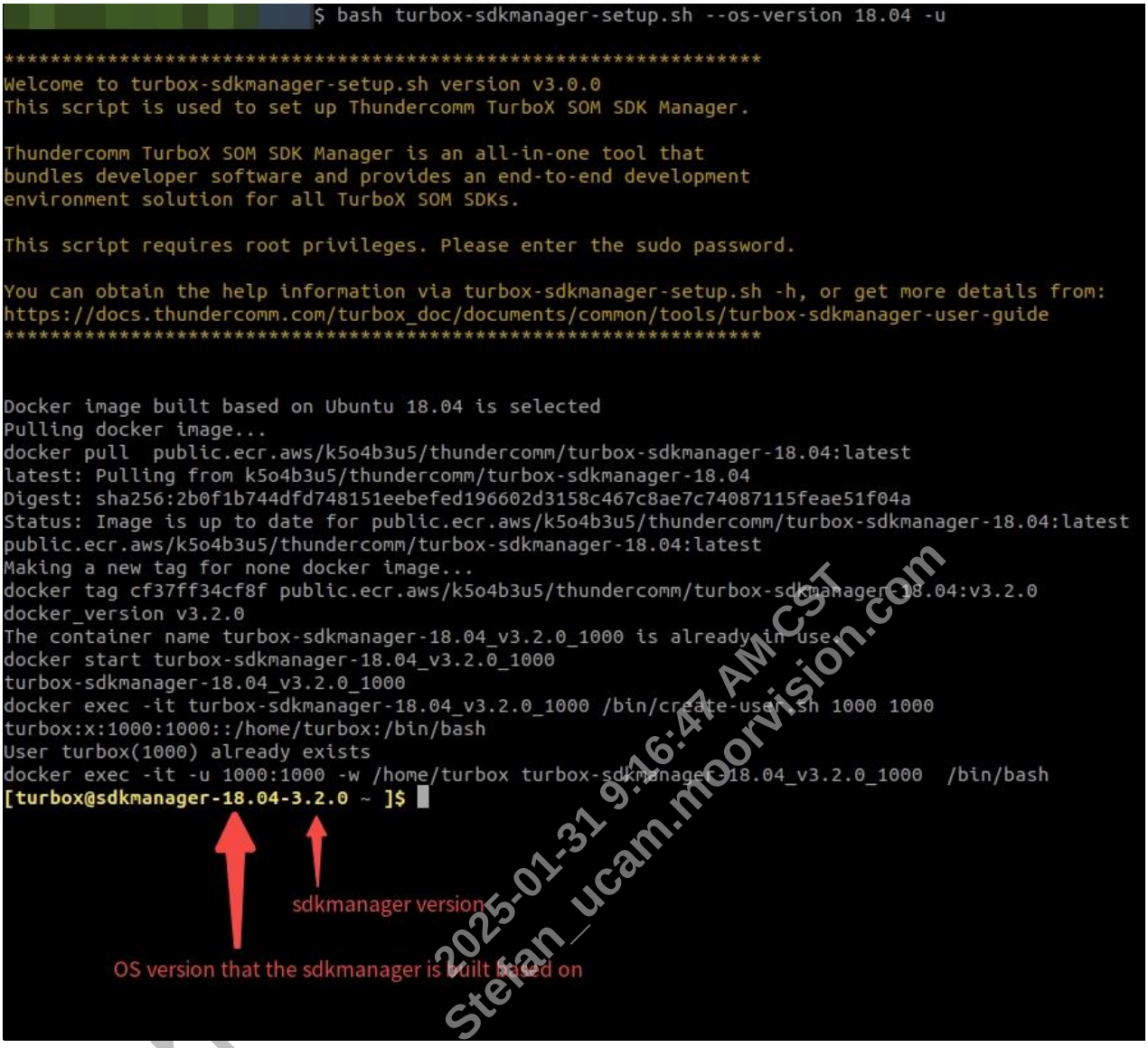


Figure 2-1.sdkmanager Window

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Chapter 3. Get Started with SDK

3.1. SDK information

Table 3-1 presents the LA SDK information, including the SDK version.

Table 3-1. LA SDK information

SDK Version	Baseline	Baseline Distribution	OS Version	SECURITY_PATCH
LA2.0.Post-CS2.r002003	QCS610.LA.2.0	Post-CS[7] 0.0.015.3	Android 12.00	2023-07-01

3.2. SDK block diagram

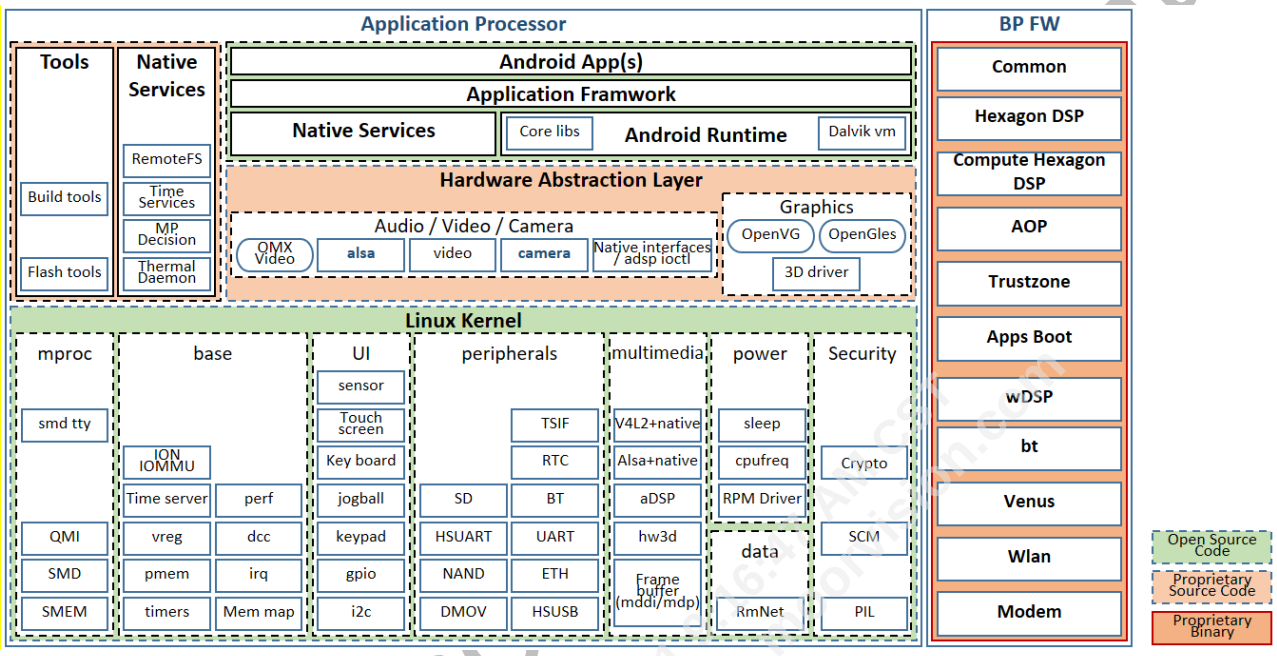


Figure 3-1.Block Diagram

3.3. SDK source tree structure

```
SDK.Turbox_C610.LA.2.0/
├── about.html -> common/about.html          //SDK version info(Product,
Distribution,Components, …)
├── contents.xml -> common/contents.xml      //image contents file
├── common/config/emmc/partition.xml         //eMMC partition file
├── LINUX
│   └── android
│       ├── bionic
│       ├── bootable/bootloader/edk2         //bootloader sources
│       ├── bootable/recovery                //recovery sources
│       ├── build             //basic config files for development kit, such as files for
system build rules and generis
│       ├── dalvik
│       ├── developers              //developer directory
│       ├── development             //infrastructure-related files for Android application
development
│       └── device                       //device related config
│           └── qcom/sm6150              //QCS610 device-related config directory
│       ├── external                     //files related to open-source development kit
│       ├── frameworks                   //source code under Android frameworks
│       ├── hardware/qcom                //code for QCS610 hardware abstraction layer
 (bootctrl, Display, gps, media, vr, …)
```

```
│       ├── kernel
│       │   └── msm-5.4                                  //kernel sources
│       │       ├──arch/arm64/boot/dts/vendor/qcom       //DT source files
│       │       └──arch/arm64/configs/vendor             //kernel config files
│       ├── libcore                                      //core library files
│       ├── libnativehelper          //dynamic library, which is the basis to JNI library
│       ├── Makefile                                  //global Makefile to define build rules
│       ├── packages                                  //application packages
│       ├── pdk                                       //plug development kit
│       ├── platform_testing                          //platform testing files
│       ├── prebuilts                //prebuild resources under x86 and arm framework
│       ├── sdk                                       //sdk and simulator
│       ├── system                   //underlying file system, application, and component
│       ├── test                                      //Android Vendor testing framework
│       ├── toolchain                                 //tool chain files
│       ├── tools                                     //tools files
│       └── vendor                                    //QCOM private repository
│           └── qcom
│               ├── opensource
│               └── proprietary
├── turbox_build.sh -> turbox/tools/build_script/common/turbox_build.sh      //SDK build
file.
├── turbox_flash.sh -> turbox/tools/build_script/device/turbox_flash.sh      //Fastboot
flash tool
├── turbox
│   └── tools
│       ├── build_script         //SDK build script(android build, zip, …)
│       └── flash_tools          //Flash tools files(qdl, …)
├── adsp_proc             //aDSP image is released as a binary and no build compilation is
needed
├── aop_proc              //AOP image is released as a binary and no build compilation is
needed
├── boot_images           //BOOT image is released as a binary and no build compilation is
needed
├── btfm_proc             //BTFM image is released as a binary and no build compilation is
needed
├── btfm_proc_CMC         //BTFM_CMC image is released as a binary and no build
compilation is needed
├── cdsp_proc             //cDSP image is released as a binary and no build compilation is
needed
├── modem_proc            //MODEM image is released as a binary and no build compilation
is needed
├── trustzone_images      //TZ image is released as a binary and no build compilation is
needed
├── venus_proc            //VIDEO image is released as a binary and no build compilation
is needed
├── wdsp_proc             //wDSP image is released as a binary and no build compilation is
needed.
└── wlan_proc             //WLAN image is released as a binary and no build compilation is
needed.
```

3.4. Download SDK and image package

Follow the steps below to initialize sdkmanager, download the SDK code and download the image package provided by Thundercomm.

➲ **NOTE:** Please download the SDK code and image package in the sdkmanager.

3.4.1. Initialize sdkmanager

Run `sdkmanager --init` to initialize sdkmanager. Refer to *TurboX SOM SDK Manager User Guide* for details.

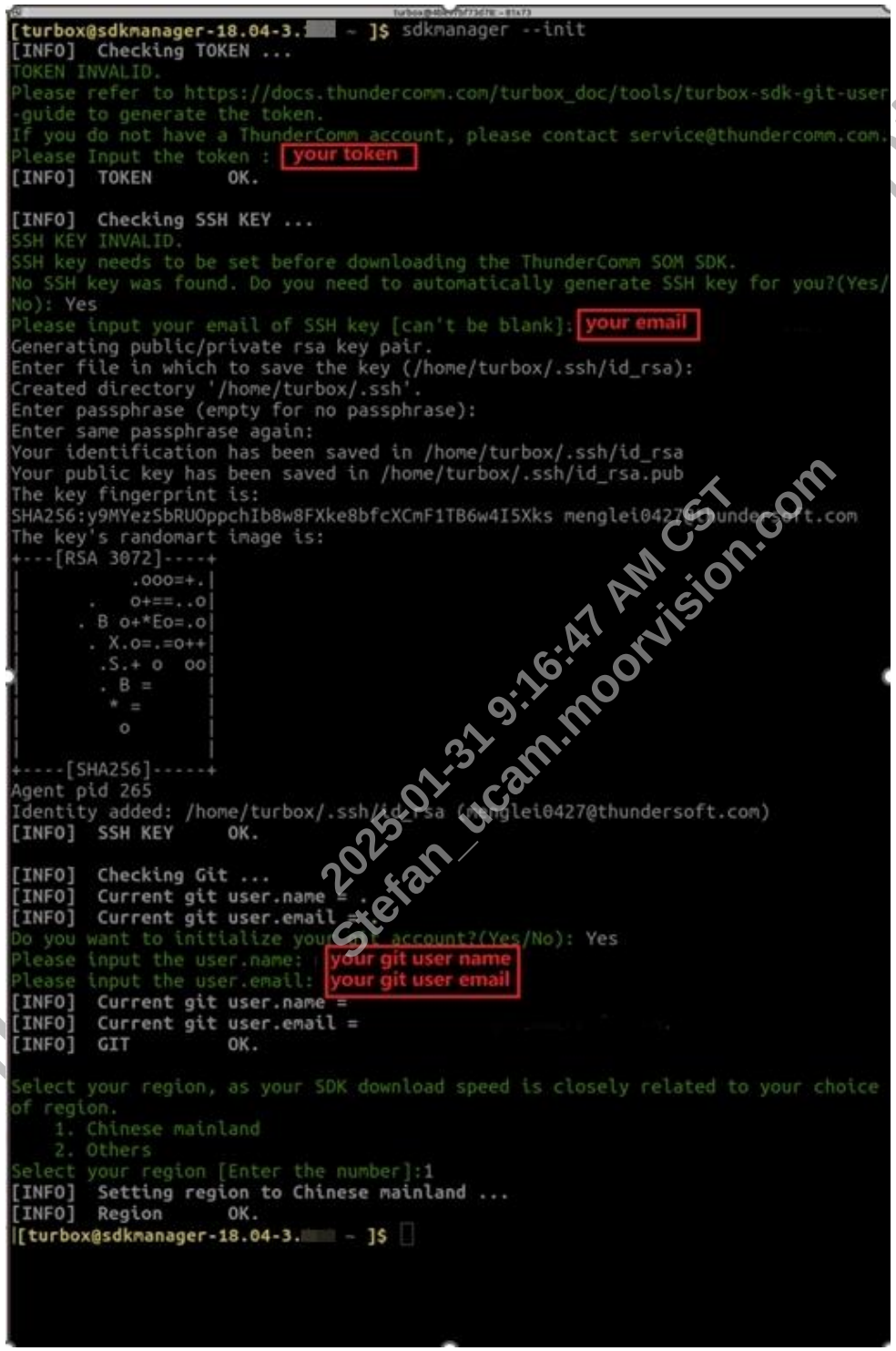


Figure 3-2.sdkmanager Initialization Windows

➲ **NOTE:** As shown in Figure 3-2, personal account information is hidden.

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3.4.2. Download SDK

Run `sdkmanager --sdk_download` to download the SDK code. Refer to *TurboX SOM SDK Manager User Guide* for more details.

Step 1.Enter "**21**" to select the product: TurboX_C610_C410.

Step 2.Enter "**1**" to select the SDK branch: SDK.Turbox-QCS610.LA.2.0.

Step 3.Enter "**5**" to select the version: turbox-c610-la2.0-dev.release.Post-CS2.r002003.



Figure 3-3.sdkmanager SDK Download Windows

➲ NOTES:

- In the step of "`Product choices are as follows:`", as the product number varies with the account privilege, you need to enter the corresponding product number according to your actual screen display.
- When defining the "`<relative path>`", please note that this path directory can only be one level; otherwise, it may cause compilation errors.

3.4.3. Download image package

Run `sdkmanager --image_download` to download the image package, refer to *TurboX SOM SDK Manager User Guide* for more details.

Step 1. Enter "**21**" to select the project: TurboX_C610_C410.

Step 2. Enter "**1**" to select the SDK image release: C610.LA2.0.Post-CS2.r002003.

Step 3. Enter "**3**" to select the SDK image package: FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip.

Alternatively, enter "**1**", "**2**" or "**4**"to select another SDK image package.



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Figure 3-4.sdkmanager Image Download Windows

➲ **NOTE:** In the step of "`Product choices are as follows:`", as the serial number varies with the account privilege, please select the correct serial number according to your screen display.

Customers can also obtain image package provided by Thundercomm from:

- **For userdebug version:**
  - *FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip*
    MD5: 066861560278080dddf844e051d8c191
  - *FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip*
    MD5: 61146ee9964452ba268243e7de168205
- **For user version:**
  - *FlatBuild_Turbox_C610_xx.xx_LA2.0.R.user.Post-CS2.r002003.zip*
    MD5: 9502668577014d3b0a98945bd10ef334
  - *FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.user.Post-CS2.r002003.zip*
    MD5: a0ec06f796d52b2b1901e9327a70352d

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Chapter 4. Build Images and Pack ROM Package

4.1. Build images

To compile code, run the following script file in the root directory of source code for building debug version.

Please run the commands in the sdkmanager.

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$ cd workspace/sourcecode/turbox-c610-la2.0-
dev.release.Post-CS2.r002003/
[turbox@sdkmanager-18.04-3.2.0 turbox-c610-la2.0-dev.release.Post-
CS2.r002003 ]$ ./turbox_build.sh -a -l -v userdebug
[turbox@sdkmanager-18.04-3.2.0 turbox-c610-la2.0-dev.release.Post-
CS2.r002003 ]$ ./turbox_build.sh -u -v userdebug
```

➲ **NOTES:**

- If you want to build user version, please change `-v userdebug` to `-v user` in the commands.
- For detailed usage method, input `./turbox_build.sh -h` to view the help information.

4.2. Pack ROM package

Follow the steps below to pack ROM package for debug version.

➲ **NOTE**: Make sure that you run the commands in the sdkmanager.

Step 1. Build image. Refer to 4.1. Build images for more details.

Step 2. Run the following command to pack the ROM package. You can find the ROM package at *turbox/output*.

- Pack **Fastboot** image

  ```
  [turbox@sdkmanager-18.04-3.2.0 turbox-c610-la2.0-dev.release.Post-
  CS2.r002003 ]$  ./turbox_build.sh --zip_fastboot_build -l -v userdebug
  ```

- Pack **Flat** image

  ```
  [turbox@sdkmanager-18.04-3.2.0 turbox-c610-la2.0-dev.release.Post-
  CS2.r002003 ]$  ./turbox_build.sh --zip_flat_build -l -v userdebug
  ```

➲ **NOTES:**

- If you want to pack user version, please change `-v userdebug` to `-v user` in the commands.
- Alternatively, customers can obtain ROM package provided by Thundercomm (refer to 3.4.3 Download image package).
- For detailed usage method, input `./turbox_build.sh -h` to view the help information.

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Chapter 5. Flash images to board

5.1. Flash FlatBuild images

➲ **NOTES**:

- FlatBuild is an image format released by Thundercomm, which can be flashed in EDL (Emergency Download) mode. When a device is unable to boot due to software issues, you can repair it by flashing FlatBuild.
- You can flash images on Linux or Windows through Tflash. For tool acquisition and operation procedures, please refer to TurboX Flash User Guide.

5.1.1. Flash FlatBuild images on Linux

➲ **NOTES**:

- Make sure that you run the commands in the sdkmanager.
- Stop the ModemManager on your host PC and close the application that occupies the ttyUSB0 port, such as **Minicom**.

```
$ sudo systemctl stop ModemManager
```

Step 1. Enter the **9008 Mode**.

1) Power off the device by disconnecting the power cable and USB cable.

2) Connect the power cable.

3) Hold the **Force_USB_Boot** button.

4) Press the **Power on** button and wait for about 3 seconds.

5) Release the **Force USB Boot** button and **Power on** button.

6) Connect the Type-C USB cable to an Ubuntu computer.

To check if your device has entered the **9008 Mode**, run the following command:

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$ lsusb
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 085: ID 413c:2113 Dell Computer Corp.
Bus 001 Device 123: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
Bus 001 Device 084: ID 413c:301a Dell Computer Corp.
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
[turbox@sdkmanager-18.04-3.2.0 ~ ]$
```

Figure 5-1. Check 9008 Mode Windows

Step 2. Prepare the image package.

- To obtain Thundercomm package, please refer to 3.4.3 Download image package.
- To pack your own image package, please refer to 4.2. Pack ROM Package.

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Step 3.Run `tflash` to flash the image.

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$ cd workspace/images/
[turbox@sdkmanager-18.04-3.2.0 images ]$ ls
FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip
FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip
[turbox@sdkmanager-18.04-3.2.0 images ]$ unzip -q FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r0020
03.zip
[turbox@sdkmanager-18.04-3.2.0 images ]$ cd FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003
[turbox@sdkmanager-18.04-3.2.0 FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003 ]$ tflash

************************************************************************
Welcome to /usr/bin/tflash version 1.0.0
This script is used to flash FlatBuild and CDT to your device

This script requires root privileges. Please enter the sudo password.

You can obtain the help information via /usr/bin/tflash -h, or get more details form:
https://docs.thundercomm.com/turbox_doc/documents/common/tools/turbox_tflash-user-guide
************************************************************************


Image Search Path: /home/turbox/workspace/images/FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r00200
3
Programmer File: prog_firehose_ddr.elf
Storage Type: eMMC
Image Flash: true
CDT Flash: false
RawProgram Files: rawprogram_unsparse0.xml
Patch Files: patch0.xml
Reset: false
Erase: false
Erase All: false

Stop the ModemManager on your host PC by implementing "sudo systemctl stop ModemManager".
If you require to use the ModemManager, restart it after the flashing process is completed.
The ModemManager can be restarted with the "sudo systemctl start ModemManager" command.
Stop the ModemManager on your host PC and press Enter on your keyboard to continue...

sudo systemctl stop ModemManager
System has not been booted with systemd as init system (PID 1). Can't operate.
sudo QSaharaServer -p /dev/ttyUSB0 -s 13:prog_firehose_ddr.elf
Binary build date: Oct 21 2020 @ 11:45:40
QSAHARASERVER CALLED LIKE THIS: 'QSaharaServe'Current working dir: /home/turbox/workspace/images/FlatBuild_Tur
box_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003
Sahara mappings:
2: amss.mbn
6: apps.mbn
8: dsp1.mbn
10: dbl.mbn
11: osbl.mbn
12: dsp2.mbn
16: efs1.mbn
17: efs2.mbn
20: efs3.mbn
21: sbl1.mbn
22: sbl2.mbn
23: rpm.mbn
25: tz.mbn
28: dsp3.mbn
29: acdb.mbn
30: wdt.mbn
31: mba.mbn
13: prog_firehose_ddr.elf
Wrote to /sys/power/wake_lock
Requested ID 13, file: "prog_firehose_ddr.elf"
651812 bytes transferred in 0.190564 seconds (3.2620MBps)


File transferred successfully
```

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Figure 5-2. Flash the Image *Windows*

If the following log appears, it indicates that the flashing progress is successful.

```
14:11:32: INFO:                 _             (done)
14:11:32: INFO:     | |
14:11:32: INFO:   __| | ___  _ __   ___
14:11:32: INFO:  / _` |/ _ \| '_ \ / _ \
14:11:32: INFO: | (_| | (_) | | | |  __/
14:11:32: INFO:  \__,_|\___/|_| |_|\___|
14:11:32: INFO: {All Finished Successfully}

14:11:32: INFO: Overall to target 63.395 seconds (37.36 MBps)
14:11:32: INFO: {percent files transferred 100.00%}

Writing log to '/home/turbox/workspace/images/FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003/p
ort_trace.txt', might take a minute


Log is '/home/turbox/workspace/images/FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003/port_trac
e.txt'
```

Figure 5-3. Flash the Image *Windows*

Step 4.Unplug and plug in the power cable to restart the device.

5.1.2. Flash FlatBuild images on Windows

Step 1. Enter the **9008 Mode**.

1) Power off the device by disconnecting the power cable and USB cable.

2) Connect the power cable.

3) Hold the **Force_USB_Boot** button.

4) Press the **Power on** button and wait for about 3 seconds.

5) Release the **Force USB Boot** button and **Power on** button.

6) Connect the Type-C USB cable to a Windows PC.

To check if your device has entered the **9008 Mode**, view the port number in "Device Manager" of your Windows PC (as shown below):

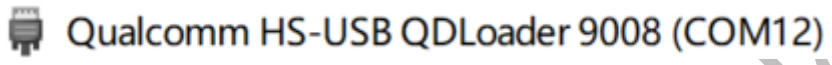


Figure 5-4. Check 9008 Mode Windows

Step 2. Prepare the image package and extract it to the local directory.

- To obtain Thundercomm package, please refer to 3.4.3 Download image package.
- To pack your own image package, please refer to 4.2. Pack ROM Package.

Step 3. Flash FlatBuild images with 9008.

1) Storage Type needs to select `EMMC`.

2) Click on UpdatePort to select `Qualcomm HS-USB QDLoader 9008 (COM*)`.

3) Click on Browse, locate the local `FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003` folder and select it, then select the files `prog_firehose_ddr.elf`, `rawprogram_unsparse0.xml`, and `patch0.xml` in sequence.

4) Click on download to start flashing.

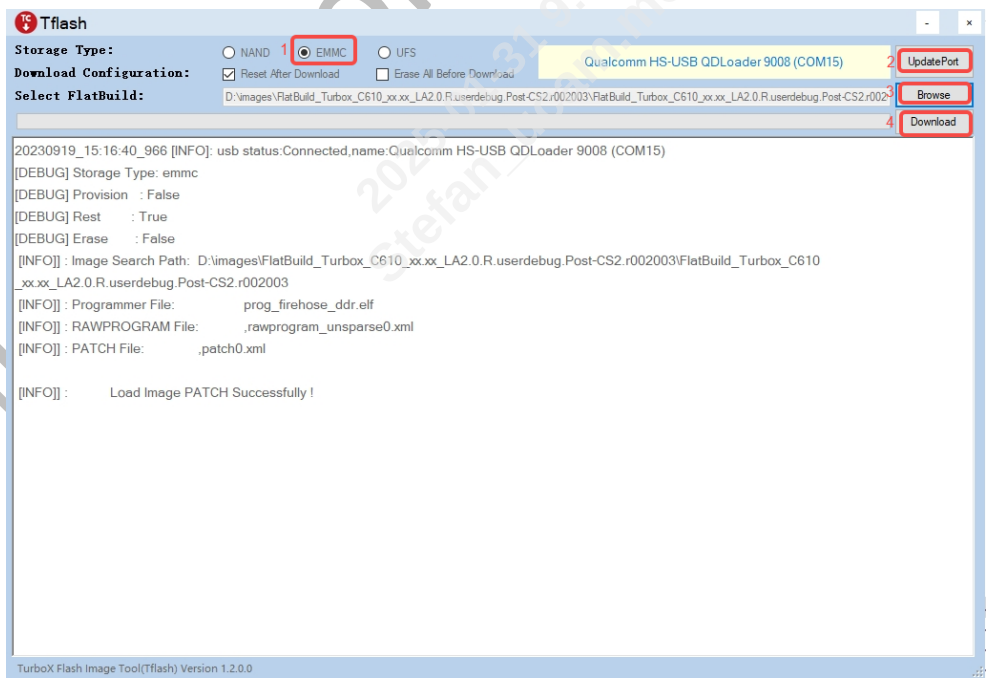


Figure 5-5. Flash the Image Windows

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If the following log appears, it indicates that the flashing progress is successful.

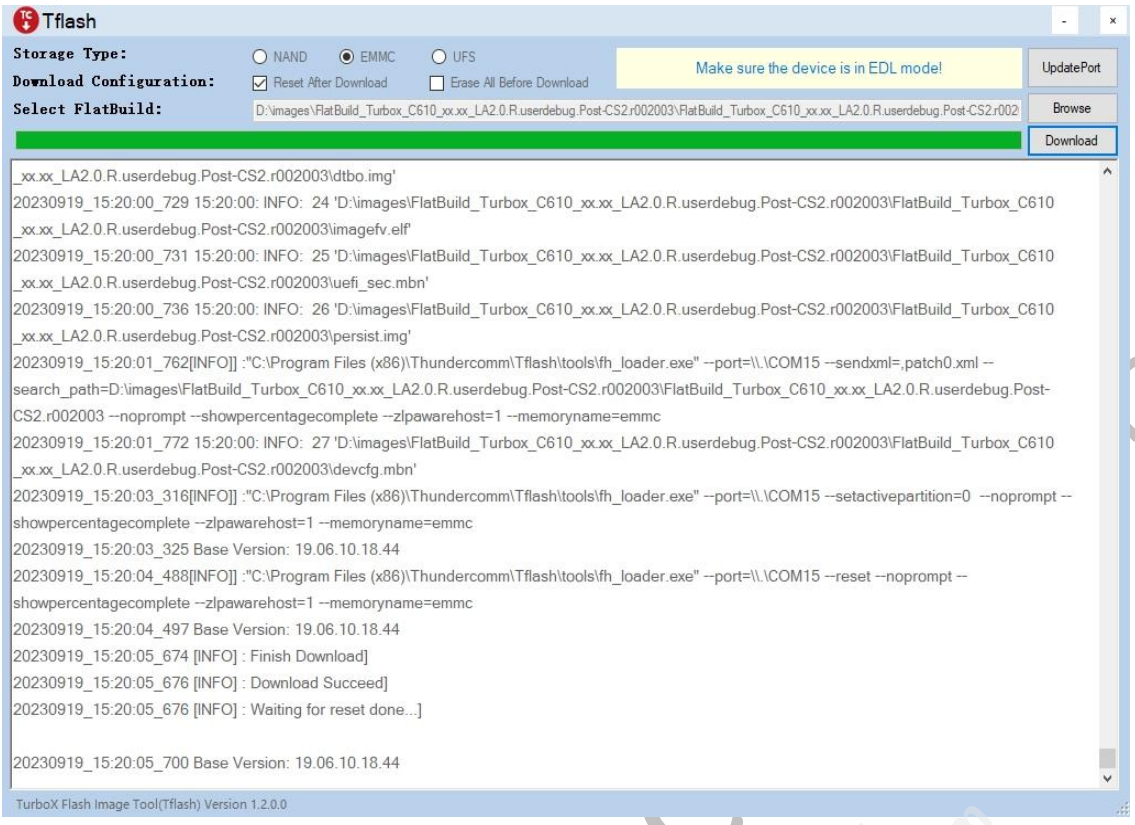


Figure 5-6.Flash the Image Windows

5.2. Flash Fastboot images

➲ **NOTES**:

- Fastboot is an image format released by Thundercomm, which can be flashed in Fastboot mode.
- The device needs to be able to enter Fastboot mode in normal booting condition through commands or button pressing.
- If the device fails to boot up after image flash with **Fastboot**, try to flash the **FlatBuild** image again with Tflash.

Follow the steps below to flash Fastboot images on Linux.

Step 1.Enter the **Fastboot** mode.

Option 1.Enter the following commands when the device is in **ADB** mode.

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$  adb root
[turbox@sdkmanager-18.04-3.2.0 ~ ]$  adb reboot bootloader
```

Option 2.Power off the device, hold the **Volume down** button and press the **Power on** button, then release the **Volume down** button about 5 seconds later.

➲ **NOTE**:

To check if the device has entered the **FastBoot** mode, run the following command:

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$  fastboot devices
```

The progress is successful with the following output displayed on the shell:

```
21f10525     fastboot
```

Step 2.Run `sudo ./turbox_flash.sh --all -r -f -v userdebug` to flash the image.

Execute the commands below to flash images to the board. Replace **<folder name of image package>** with the actual image package name of your product.

```
[turbox@sdkmanager-18.04-3.2.0 ~ ]$ cd workspace/images/
[turbox@sdkmanager-18.04-3.2.0 images ]$ unzip -q FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r0
02003.zip
[turbox@sdkmanager-18.04-3.2.0 images ]$ ls
FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003
FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip
FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003
FlatBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003.zip
[turbox@sdkmanager-18.04-3.2.0 images ]$ cd FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003
[turbox@sdkmanager-18.04-3.2.0 FastbootBuild_Turbox_C610_xx.xx_LA2.0.R.userdebug.Post-CS2.r002003 ]$ sudo ./tur
box_flash.sh --all -r -f -v userdebug
target reported max download size of 805306368 bytes
sending 'partition:0' (33 KB)...
OKAY [  0.009s]
writing 'partition:0'...
OKAY [  0.774s]
finished. total time: 0.783s
target reported max download size of 805306368 bytes
sending 'bluetooth_a' (876 KB)...
OKAY [  0.027s]
writing 'bluetooth_a'...
OKAY [  0.008s]
finished. total time: 0.035s
target reported max download size of 805306368 bytes
sending 'cmnlib64_a' (507 KB)...
OKAY [  0.019s]
writing 'cmnlib64_a'...
OKAY [  0.006s]
finished. total time: 0.025s
target reported max download size of 805306368 bytes
sending 'cmnlib_a' (389 KB)...
OKAY [  0.020s]
writing 'cmnlib_a'...
OKAY [  0.005s]
```

Figure 5-7.Flash the Image Windows

➲ **NOTES**:

- If you want to flash user version, please change `-v userdebug` to `-v user` in the commands.
- Enter `./turbox_flash.sh -h` to acquire detailed usage.

If the following log appears, it indicates that the flashing progress is successful.

```
OKAY [  0.055s]
sending 'userdata' (6996 KB)...
OKAY [  0.173s]
writing 'userdata'...
OKAY [  0.000s]
finished. total time: 0.228s
Loading complete, Device rebooting ...
rebooting...

finished. total time: 0.050s
```

Figure 5-8.Flash the Image Windows

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Chapter 6. Features

This chapter introduces the software features supported by the DK.

6.1. Presented features

This section describes the key software features that have been included in earlier versions.

- **Kernel**
  - Linux Kernel 5.4
  - 64-bit only
- **OS**
  - Android 12
  - F2FS + FBE
  - Dynamic partition
  - Fastboot support
  - Boot image header version 1
  - Camera 2.0 (HAL 3.4)
  - Keymaster 4.0
  - Vulkan 1.1
  - OTA Android S to S
  - Incremental OTA
- **CV**
  - FastCV
  - TF-Lite, OpenGL, SNPE, QNN verification
- **Security**
  - Secure boot
  - Secure video encoding
  - Secure camera
  - Secure DSP
- **Connectivity**
  - WLAN 802.11ac
  - USB 3.1 Gen 1
  - UVC/UAC host support
  - Bluetooth 5.0
- **Storage**
  - eMMC
  - Dual rank2 channel LPDDR4x memory support
- **Multimedia**
  - YUV + Bayer capture
  - sHDR v3.8
  - sHDR tuning through Chromatix
  - LDC
  - TNR (MCTF)
  - LT6911UXC HDMI-IN

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6.2. New features

The following bug has been fixed in this SDK version:

- IMX415 module camera cannot take photos or record videos.

6.3. On-going features

None.

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Chapter 7. Test Report

➲ **NOTE:** Refer to Table 7-1 for the test results on SKU3 (QC610 4GB + 64GB).

Table 7-1. Sanity test report

Test case	Status
Fastboot flash	Pass
QFIL Flat flash	Pass
QFIL Meta flash	Pass
Recovery mode (adb & volume up)	Pass
Retrieve the abl/xbl/kernel version	Pass
Power on button	Pass
Volume up/down button	Pass
Force USB Boot button	Pass
Fastboot (adb & volume down)	Pass
BT/Wi-Fi/Gen LED	Pass
LSM6DSOTR sensor	Pass
SD Card auto mount/umount/multiple plug & unplug	Pass
Debug UART	Pass
Type A connection	Pass
Interaction between Type A and Type C	Pass
Type C connection	Pass
DP	Pass
LCD touch panel	Pass
MIPI LCD (ili9881c)	Pass
MIPI LCD (ili9881p)	Pass
UAC/UVC host	Pass
DMIC	Pass
Line in	Pass
Audio headphone	Pass
Audio speaker	Pass
BT connection	Pass
Wi-Fi connection	Pass
Battery	NA
POE15W power supply	NA
12V DC power supply	Pass
Charge the battery via type-c	NA
Ethernet_Network port indicator	Pass
DHCP client	Pass
Sleep mode wakeup	Pass
Random read/write	Pass

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Test case	Status
HDMI IN	Pass
IMX577 module capture & video	Pass
IMX415 module capture & video	Pass
QNN	Pass
SNPE	Pass
OpenGL	Pass
FastCV	Pass
TF-Lite	Pass

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Chapter 8. Known Issues

Table 8-1. List of known issues

<table>
<tr><th>Issue No.</th><th colspan="2">Description</th><th>Workaround</th></tr>
<tr><td rowspan="8">SOMC610-176</td><td colspan="2">IMX577 front camera only supports certain resolutions:</td><td rowspan="8">Open the camera app, switch to the rear camera, and set the resolution to "OK". Then switch to the front camera, and you will be able to take photos and record videos using the front camera.</td></tr>
<tr><td>Format</td><td>Test Result</td></tr>
<tr><td>4K</td><td>NG</td></tr>
<tr><td>1080P</td><td>NG</td></tr>
<tr><td>720P</td><td>OK</td></tr>
<tr><td>480P</td><td>OK</td></tr>
<tr><td>VGA</td><td>NG</td></tr>
<tr><td>CIF</td><td>OK</td></tr>
<tr><td>QVGA</td><td>OK</td></tr>
<tr><td>SOMC610-165</td><td colspan="2">Connecting HDMI-IN during startup will cause the system to crash.</td><td>Do not connect HDMI-IN during startup; connect it after the system has booted up.</td></tr>
</table>

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